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Short Communication

Investigation on Raman spectral features of a coated tablet under variation of its orientation respective to laser illumination and measurement of nominal coating thickness of packed tablets

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ABSTRACT

To investigate Raman spectral features of a coated biconvex tablet under variation of its orientation respective to laser illumination, spectra of the tablet were collected by illuminating laser on 12 different locations on the tablet with 3 different illumination angles of 45, 75 and 90°. The spectral variations were more substantial when the tablet faces with engraved letters and greater surface curvature were measured, since the sampled volume of coating relative to that of a core tablet changed significantly under these circumstances as the illumination angle varied. The preliminary examination confirmed that the acquisition of tablet-representative spectra was the requisite for reliable measurement of coating thickness. Then, to mimic real monitoring of coating process, Raman spectra were directly collected on a packing of 30 tablets with repetition of random tablet packing up to 15 times and univariate models utilizing the intensity of coating peak at 638 cm^{-1} were developed using the cumulatively averaged spectra with an average weight of the 30 tablets as a reference. To acquire less tablet orientation-sensitive spectra, a wide area illumination (WAI) scheme providing a large sampling area (28.3 mm^2) on a tablet with a long focal length (~25 cm) was employed. The averaging of the first to seventh spectra, equivalently utilizing more packing-representative spectra for quantitative analysis, made the measurement of nominal coating thickness of packed tablets accurate.

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1. Introduction

Raman and near-infrared (NIR) spectroscopy have been extensively studied as a potent member of process analytical technology (PAT) for fast and non-destructive quality control of diverse samples [1,2]. Both methods have their own advantages and disadvantages as a PAT tool; while, Raman spectroscopy able to provide more detailed and specific structural information of a sample with a narrower bandwidth could be a more preferable choice, if a sample does not seriously emanate fluorescence. Due to the capability, it has been frequently employed for diverse analyses of pharmaceutical samples, such as monitoring of tablet coating [3], determination of crystallinity [4] and measurement of active pharmaceutical ingredient (API) concentrations [5,6].

Once spectroscopic monitoring of tablet coating process is considered, along with Raman spectroscopy, NIR spectroscopy and terahertz pulsed imaging (TPI) could be potential alternatives. In the case of using NIR spectroscopy, it would additionally allow to measure moisture content of tablets; however, NIR bands are much less specific and more overlapping with each other compared to Raman bands, thereby making the construction of a robust multivariate model more challenging and uneasy. TPI has important advantages in the determination of coating thickness, such as a higher axial resolution and ability of examining multilayer complex coating; therefore, a considerable number of related studied have been reported [7]. However, several hurdles to overcome for the adoption of TPI in routine analysis exist, such as spectral sensitivity to the distance as well as orientation of measuring tablets and generation of potential artifacts in TPI spectra due to the nature of time-domain based data acquisition.

When Raman spectroscopy is employed for the same purpose, acquired peak intensities of a coating material must vary quantitatively in accordance with the degree of coating on a tablet

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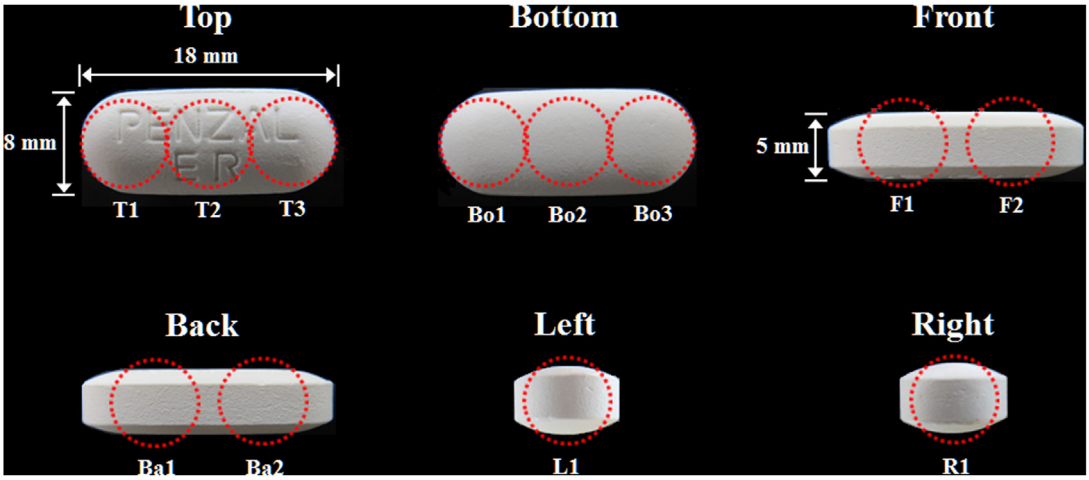

Fig 1. Pictures of a biconvex tablet used in this study. The dotted circles indicate the laser illumination spots with actual size (6 mm in diameter). T, Bo, F, Ba, L and R stand for the laser illuminations on top, bottom, front, back, left and right of the tablet.

[8–10]. However, in some cases, Raman spectra collected from a coated tablet may not be representative of actual amount of coated material. For example, peak intensities of coating material on a non-spherical tablet could vary as orientations of the tablet vary relative to direction of laser illumination. Under this circumstance, the ratio between sampled volumes of coating and a core tablet varies depends on the illumination angle; therefore, the intensities of coating would not be consistent even though the same tablet is measured. In addition, the sample surface is difficult to be located reproducibly at a laser focal point under this situation, so it is another source that results in inconsistent peak intensity. These are obviously unwanted occurrences degrading accuracy in the monitoring of tablet coating process. Since the shapes of pharmaceutical tablets are generally non-spherical, the pointed issues need to be seriously considered for Raman measurement.

For the examination, Raman spectra of a coated tablet (biconvex shape) were collected by illuminating laser on 12 different locations on the tablet as shown in Fig. 1 with 3 different illumination angles of 45, 75 and 90° to the tablet surface, and the variations of spectral features were systematically investigated. A wide area illumination (WAI) scheme able to cover a large area (28.3 mm^2) on a tablet with a long focal length (~25 cm) was employed in this study [1,11], since it could be helpful to acquire relatively less orientation-sensitive spectra owing to its capabilities of large-area sampling and less sensitivity of sample placement with regard to the focal plane.

Next, univariate models utilizing either peak intensity ratio or intensity of coating peak at 638 cm^{-1} were developed to measure tablet coating thicknesses. First, 10 individual tablets were obtained at different stages of coating process and Raman spectra of each tablet were collected with the perpendicular illumination on T2 (refer to Fig. 1). A model was developed using the spectra to correlate with an average weight of 10 tablets. Second, to mimic real monitoring of coating process, Raman spectra were directly collected on the packing of 30 tablets with repetition of random tablet packing up to 15 times and variation of accuracies, according to the sequential use of cumulatively averaged spectra for the same analysis with an average weight of 30 tablets as a reference, was evaluated. Finally, the results in these 2 different cases were comparatively discussed.

2. Experimental section

Raman spectra were collected using a wide area illumination (WAI) scheme (excitation wavelength: 785 nm) able to provide a large sampling area (28.3 mm^2) with a laser illumination over 6 mm in diameter for more representative sampling (Kaiser Optical Inc., Ann Arbor, MI, USA). For spectral acquisition, a tablet was simply positioned at the focal point of the WAI scheme. Generated Raman photons were collected by an array of 50 optical fibers and then dispersed via a holographic transmission grating to a CCD detector. The spots (dotted circles) where laser was illuminated on the tablet were depicted in Fig. 1. The laser power at the illuminated surface was 400 mW and exposure time was 3 s. At each location, Raman spectra were collected by irradiating laser with 3 different angles (90, 75, 45°) relative to the tablet surface. Pretreatments of spectra (baseline correction and area normalization) and principal component analysis (PCA) were performed using MATLAB R2007a (MathWorks Inc., Natick, MA, USA). A MATLAB toolbox for PCA was employed in this study.

The biconvex tablet with 5 mm in height and 18 mm in diameter was employed in this study as shown in Fig. 1. It was composed of mainly acetaminophen, pregelatinized starch, croscamellose sodium and hydroxy propyl methyl cellulose. To prepare calibration samples to build univariate models, tablets were periodically sampled 14 times with intervals of 30 min during tablet coating. For preparation of validation samples, tablets were separately collected 12 times during another batch of tablet coating. The weights of 10 core tablets varied from 0.7490 to 0.7730 g with average and standard deviation (SD) of 0.7619 and 0.0073 g, respectively. Tablet weight was measured using a high precision balance (XS205 Dual Range, Mettler Toledo, Columbus, OH, USA). The coating material used in this study was OPADRY® WHITE (Colorcon, Harleysville, PA, USA), composed of 62.50% of hydroxy propyl methyl cellulose, 31.25% of TiO_2 and 6.25% of polyethylene glycol 400 (PEG 400).

3. Results and discussion

3.1. Investigation of Raman spectral features of a coated tablet under its orientation change

Since the tablet studied here was biconvex in shape as shown in Fig. 1, Raman spectral features of a coated tablet would vary depends on the locations of laser illumination as well as illumination angles, even with the use of WAI scheme for spectral acquisition. Under this circumstance, intensities of coating peaks would vary since a sampled volume of coating relative to that of core tablet expects to vary. For the examination, Raman spectra were acquired at 12 different locations on a coated tablet (coated over 6 h) as described in Fig. 1. The dotted circles indicate the laser

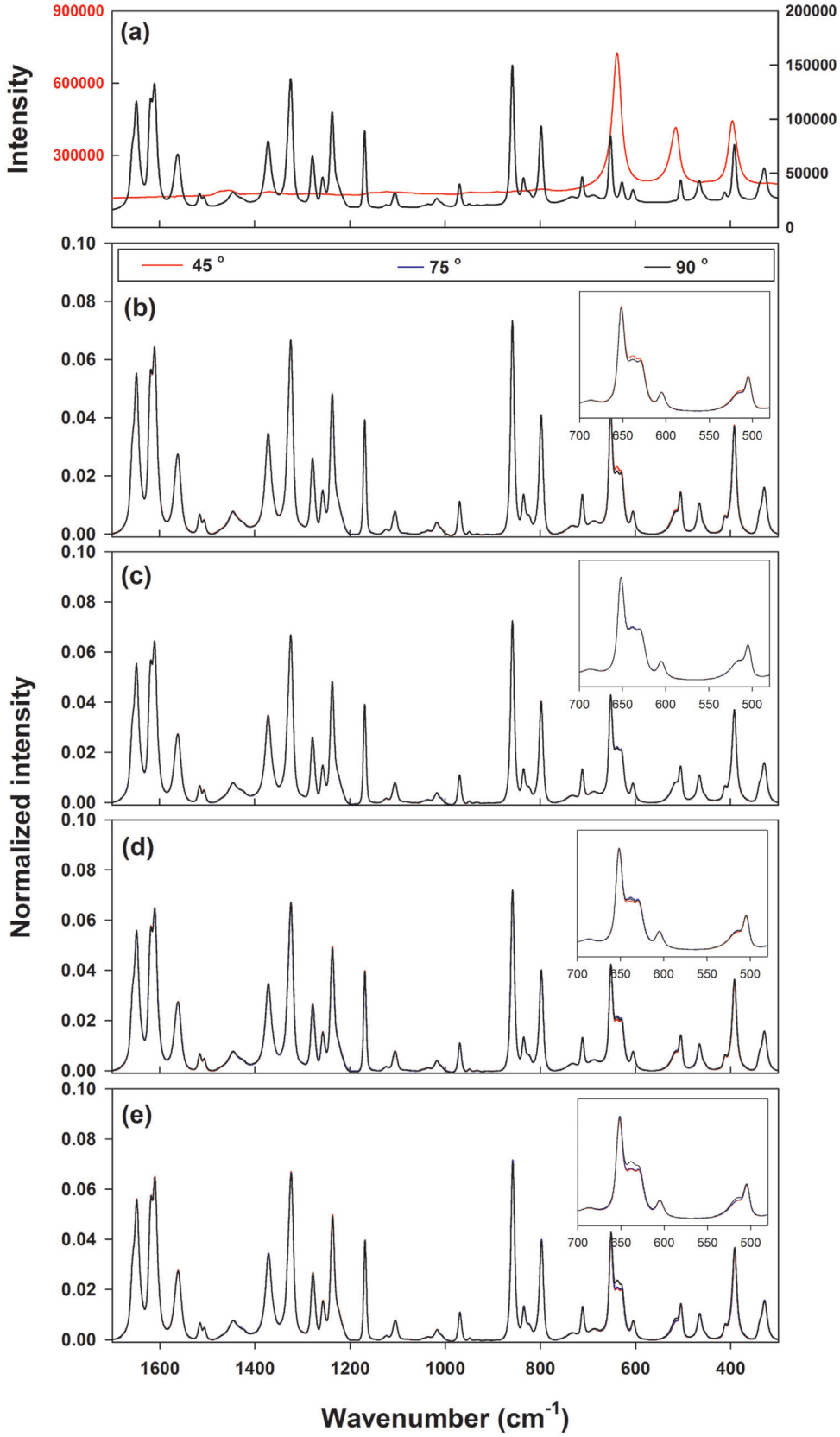

Fig. 2. Raman spectra of a pure coating material (red) and a core tablet (black) (a). Normalized Raman spectra of a coated tablet acquired at T2 (b), Bo2 (c), F2 (d) and L1 (e) with 3 different laser illumination angles are shown. The bands in the 700–450 cm^{-1} range are highlighted in the inset. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

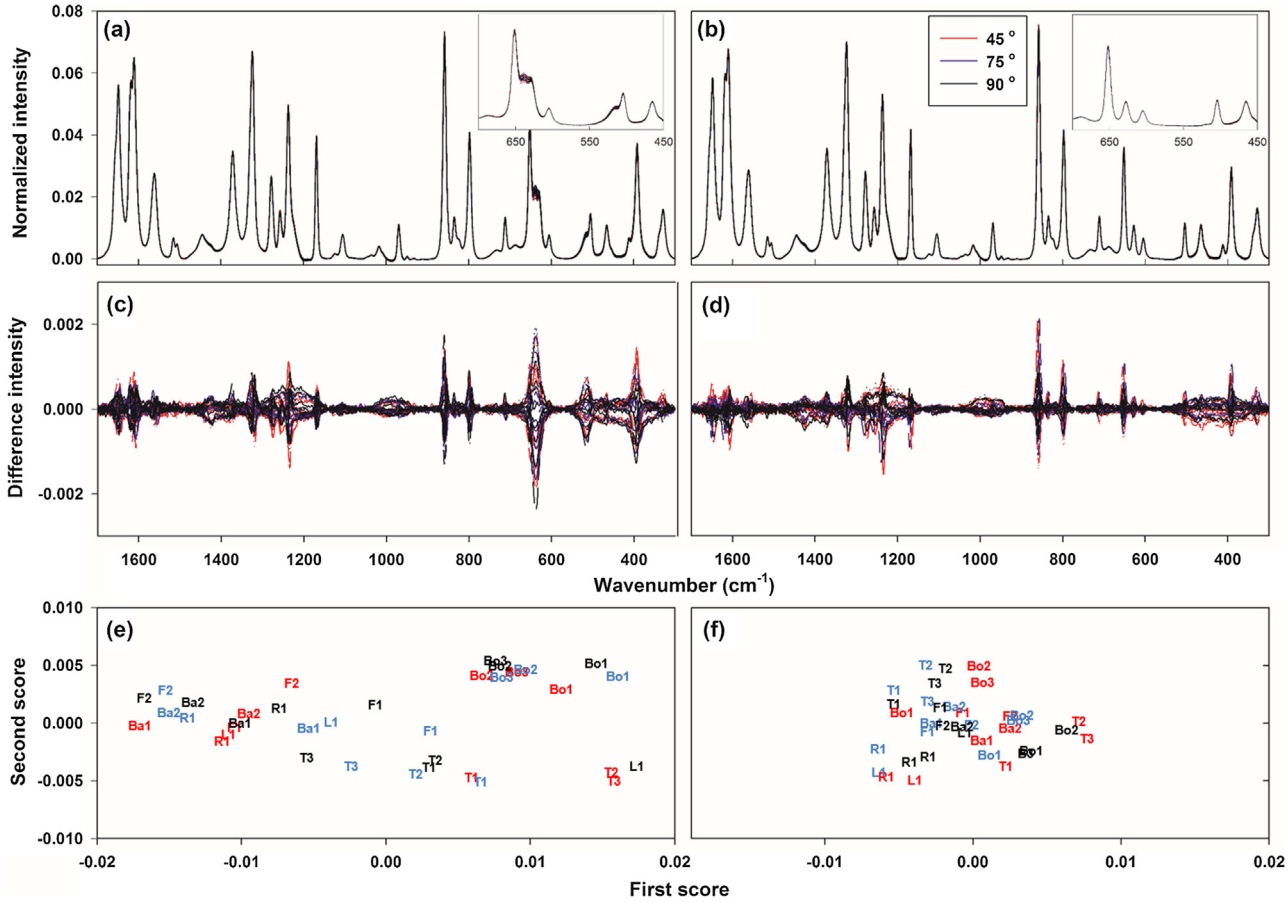

Fig. 3. All 36 normalized spectra of a coated tablet collected at 12 locations with 3 different illumination angles (a) and difference spectra obtained by subtracting each spectrum from the average spectrum of all 36 spectra (c). Normalized spectra of a core tablet (b) and the corresponding difference spectra (d) are also shown for comparison. The inset magnifies the peaks in the 700–450 cm^{-1} range. Score scatter plots (the first vs. second scores) construed using the spectra (700–300 cm^{-1} range) of coated (e) and uncoated tablets (f) are also shown. The letters indicates the sampled locations, and black, blue and red colors correspond to the 90°, 75° and 45° illuminations, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

illumination spots with actual size (6 mm in diameter). T, Bo, F, Ba, L and R stand for the laser illuminations on top, bottom, front, back, left and right of the tablet, respectively. The letters (PENZAL ER) were engraved at the top (T). At each location, laser was irradiated with 3 different angles (90, 75, 45°) relative to the tablet surface. In the cases of F and Ba, the laser illumination area was larger than the area of tablet surface, and the surface curvature was greatest at L and R.

Fig. 2(a) shows Raman spectra of a pure coating material (red) and a core tablet (black) measured at T2. The 3 major coating peaks are located at 638, 510 and 400 cm^{-1}; while, no significant peaks are observed in the 1700–700 cm^{-1} range. Fig. 2(b) and (c) shows normalized Raman spectra of the coated tablet acquired at T2 and Bo2 with 3 laser illumination angles (90°, 75° and 45°). The bands in the 700–450 cm^{-1} range are also highlighted in the inset. For the normalization, the baselines of raw spectra were corrected at 1700, 1490, 1140, 1070, 770, 570 and 300 cm^{-1} to zero such as using a continuing two-point linear baseline correction. There were no significant peaks at the selected wavenumbers. Then, intensities of the baseline-corrected spectra at each number were divided by the peak area under the 1700–300 cm^{-1} range to produce normalized spectra. Overall spectral features were similar in each case; while, the intensities of coating peak at 638 cm^{-1} fairly varied. Notably, the 45° illumination produced the greater intensity at 638 cm^{-1} in both cases since the diagonally illuminated laser radiation made the coated material more sampled compared to the perpendicular illumination. Interestingly, the 45° T2 measurement resulted in the stronger intensity at 638 cm^{-1} compared to that of 45° Bo2 measurement, since the coating material entered inside the engraved letters during tablet coating and relatively a larger amount of coating material was sampled.

Fig. 2(d) and (e) shows normalized Raman spectra of the coated tablet acquired at F2 and L1 with 3 illumination angles. In the measurement at F2, in which the sample area was smaller than the illumination area, the peak intensities at 638 cm^{-1} varied moderately as the illumination angle changed and the largest intensity variation was observed in the measurement at L1. Since the radius of curvature at L1 is greater than those of other locations, the sampled amount of the outer coating varied more substantially with the illumination angles.

Fig. 3(a) and (c) shows all 36 normalized spectra of the coated tablet collected at 12 locations with 3 different illumination angles and difference spectra obtained by subtracting each spectrum from the average spectrum of all 36 spectra, respectively. The 700–450 cm^{-1} range encompassing the coating peaks is featured in the inset. The spectral features in the 1700–700 cm^{-1} range containing no peaks of the coating material were not significantly different with each other; while, much larger variation was observed in the 700–450 cm^{-1} range, especially around 638 cm^{-1} as clearly observable in the difference spectra. RSD of the intensities at 638 cm^{-1} was 5.5%. The result manifested that the variation of sampled spots and illumination angles produced the spectra with inconsistent quantitative information on the coating, although the same tablet was measured.

In a meanwhile, Raman spectra of a core tablet were also collected with the same fashion for comparison. Fig. 3(b) and (d) shows all 36 normalized spectra of the core tablet and difference spectra obtained by subtracting each spectrum from the average spectrum of all 36 spectra, respectively. The inset magnifies the peaks in the 700–450 cm^{-1} range. The spectral features of all 36 spectra are nearly analogous without any systematic variation according to the either sampled locations or illumination angles. Relative standard deviation (RSD) of the band intensities at 858 cm^{-1} (the most intense peak) was only 1.5%. In the case of core tablet, the consistent spectral features were able to obtain after normalization, although the sampled locations and illumination angles varied in the spectral acquisition.

To alternatively visualize the variations in the spectral features of the coated tablet, principal component analysis (PCA) was separately performed using the normalized spectra of the coated (Fig. 3(a)) and uncoated tablet (Fig. 3(c)), and the distributions of obtained scores were examined. The 700–300 cm^{-1} range containing the coating peaks was used for PCA. Fig. 3(e) and (f) show the score scatter plots (the first vs. second scores) construed using the spectra of the coated and uncoated tablet, respectively. The letters indicates the sampled locations, and black, blue and red colors correspond to the 90°, 75° and 45° illuminations, respectively. As expected, the scores were much more spread out in the measurement of coated tablet due to the greater variation in the intensities of coating peaks. Especially, the scores of measurements on the top of coated tablet (T1, T2 and T3) were more greatly distributed due to the letter engraving making coating substantially more inhomogeneous. Also, the scores of L1 measurements varied largely with the illumination angles, due to the greater surface curvature at this spot as explained in Fig. 2(e). In overall, the examination of score scatter again confirms the greater fluctuation of coating peaks as the sampled locations as well as illumination angles change. Obviously, the acquisition of coating thickness-representative spectra is the requisite for reliable Raman spectroscopic monitoring of tablet coating process.

3.2. Development of quantitative models for monitoring of tablet coating

Univariate methods based on peak intensity ratio were employed to measure tablet coating thicknesses. For this purpose, the peak intensity ratio between the 858 (most intense peak of a core tablet) and 638 cm^{-1} (most intense peak of a coating material) was calculated and correlated with coating thickness. The models were developed by 2 different ways and their results were compared. First, 10 individual tablets were obtained at different stages of coating process and Raman spectra of each tablet were collected with the perpendicular illumination on T2 using the WAI scheme. Peak intensity ratios were calculated from each spectrum and correlated with an average weight of these 10 tablets (case I). Second, Raman spectra directly collected on packings of 30 tablets were used for the analysis and an average weight of 30 tablets was also used as a reference (case II). The spectral acquisition continued with repetition of random tablet packing up to 15 times and subsequent variation of accuracies was examined. The average weight, standard deviation (SD) and relative standard deviation (RSD) of core tablets were 0.7641 g, 0.0071 g and 0.93%, respectively.

Fig. 4(a) and (b) presents the plots showing correlations between the peak intensity ratios and tablet weights in the cases I and II, respectively. In the case II, the cumulatively averaged spectra of 30 tablet-packing collected with 7-time random repacking were employed. As seen, the peak intensity ratios decrease steeply at the earlier stage of coating process and their variation become less significant at the later stage. Since a back-scattering spectrum inevitably contains much greater information pertinent to components located near sample surface, amount of information on coating material relative to that of a core tablet in collected spectra become dominantly greater with the increase of coating thickness. So, the nonlinear variation of peak intensity ratio is expectable.

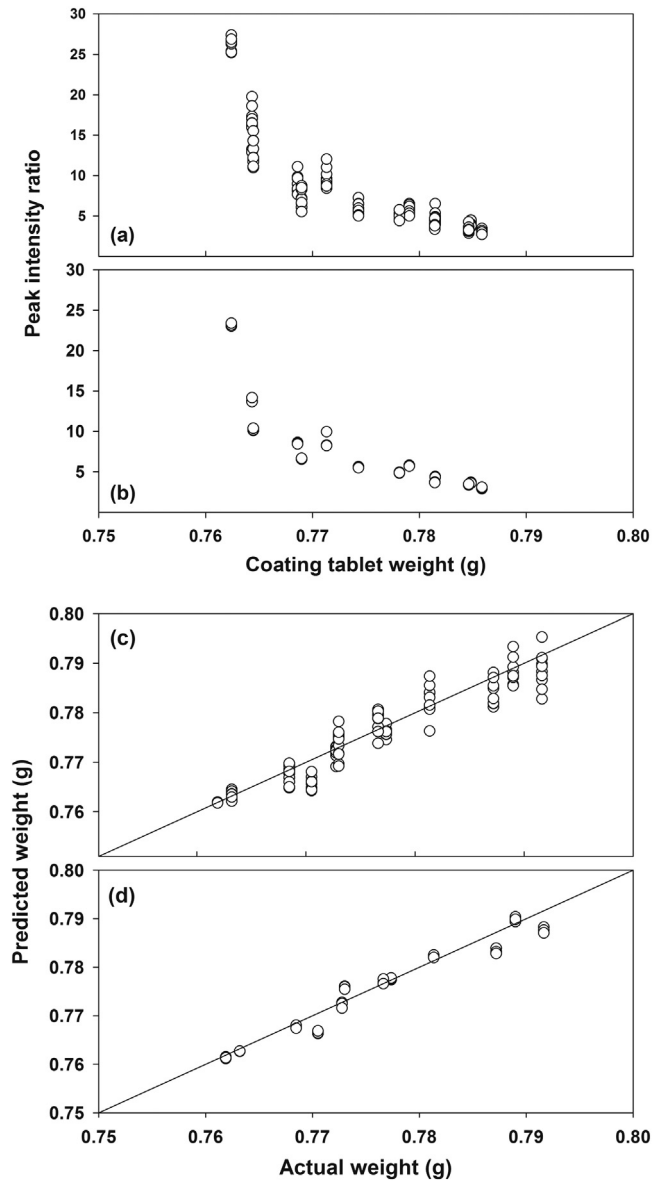

Fig. 4. Correlations of the peak intensity ratios (between the 858 and 638 cm^{-1} peak) with the corresponding tablet weights in the cases I (a) and II (b). The predicted tablet weights using the linear regression model based on correlation between the intensity of coating peak at 638 cm^{-1} and the tablet weight in the cases I (c) and II (d) are also shown.

The nonlinear correlation observed above would not be recommendable for conventional quantitative analysis; therefore, correlation between the intensity of coating peak at 638 cm^{-1} and tablet weight was examined for an alternative approach. The correlation became linear, and resulting linear regression model was used to calculate and predict coating thickness. Fig. 4(c) and (d) presents the predicted coating thicknesses vs. reference coating thicknesses in the cases I and II, respectively. As described earlier, the validation samples were separately collected during another batch of tablet coating. In the case I, the obtained standard error of calibrations (SEC) and standard error of predictions (SEP) were 0.0027 g and 0.0029 g, respectively. In the case II, when the cumulatively averaged spectra of 30 tablet-packing collected 15 times with

random repacking were consecutively used for quantitative analysis, the resulting variation of SEPs was additionally examined. The SEPs decreased by the averaging of subsequently acquired spectra, equivalently utilizing more packing-representative spectra for the analysis. After the averaging of the first to seventh spectra, there was no meaningful improvement in the accuracy. This indicated that the 7-time averaged spectrum was enough to represent the coatings of packed 30 tablets. The SEC and SEP were 0.0024 and 0.0022 g, respectively. The accuracy was improved compared to that resulted in the case I, mainly due to the use of more tablet coating-representative spectra in analysis. The spectrum collected at only T2 was insufficient to quantitatively represent the coating thickness on each tablet (Case I). In a meanwhile, the SEP was 0.0028 g when the first spectrum of the tablet packing was only used for the analysis.

4. Conclusion

Based on the observation in this study, acquisition of Raman spectra relatively less sensitive to orientations of a coated tablet respective to laser illumination was greatly demanding for reliable monitoring of coating thickness. In this context, we think that the WAI scheme was a great aid to meet the demand considerably and its long working (focal) distance was additionally advantageous to collect spectra much less sensitive to variation in tablet packings. Potential variation of particle size in a tablet that could alter intensities of relevant peaks [12,13] is an additional issue to consider for accurate determination of coating thickness. In a meanwhile, utility of Raman spectroscopy becomes weaken when a target sample emit fluorescence, usually much stronger signal than Raman signal. In this situation, the use of NIR laser decreasing the chance of fluorescence is optional; however, some of pharmaceutical components such as iron oxide (a colorant) are still NIR fluorescent substances. In final, none of three described methods (Raman, NIR and TPI) would not be an only sure method for monitoring of coating thickness in an industrial setting and further technical developments are required as described [14]. For successful applications of these methods, homogeneous coating over the surfaces of tablets providing minimal intra- and inter-tablet variations should be simultaneously accompanied.

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